KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME:

TEACHER:

MATHEMATICS

Year 8 – Practice Examination 4

Equations, Investigating Data and Mixed Topics

Practice paper for the Year 8 Yearly Examination

Instructions		
1. Attempt all questions 2. Show working where needed 3. Time Allowed 60 minutes 4. Calculators are permitted 5. Marks are shown beside each question	Equations (Section A):	/18
	Investigating Data (Section B):	/21
	Mixed Topics (Section C):	/21
	Total Marks:	/60

Section A – Equations (18 marks)

Suggested time: 20 minutes. Show all working.

Question 1 – One-step equations **(4 marks)**

Solve each equation.

a) $x + 24 = 50$ $x =$ _____ **(1 mark)**

Working: _____

b) $m - 17 = 6$ $m =$ _____ **(1 mark)**

Working: _____

c) $12p = 84$ $p =$ _____ **(1 mark)**

Working: _____

d) $y \div 8 = 5$ $y =$ _____ **(1 mark)**

Working: _____

Question 2 – Two-step equations **(4 marks)**

Solve each equation, showing your working.

a) $8x - 5 = 43$ **(2 marks)**

Working: _____

_____

_____

$x =$ _____

b) $x \div 2 + 9 = 16$ **(2 marks)**

Working: _____

_____

_____

$x =$ _____

Question 3 – Equations with a negative solution **(2 marks)**

Solve $2m + 19 = 5$.

Working: _____

_____

_____

$m =$ _____

Question 4 – Checking solutions **(2 marks)**

Use substitution to show whether $x = 3$ is the solution to $6x + 4 = 22$.

Working: _____

_____

_____

_____

Is $x = 3$ the solution? (circle one) YES / NO

Question 5 – Worded problem **(3 marks)**

I think of a number, divide it by 2, and then add 7. The answer is 19.

a) Write an equation to represent this problem. Use *n* for the number. **(1 mark)**

Equation: _____

b) Solve your equation to find the number. **(2 marks)**

Working: _____

_____

_____

n = _____

Question 6 – Worded problem **(3 marks)**

A phone plan costs $25 per month plus a $40 connection fee. Zara paid $290 altogether. Let *m* be the number of months Zara had the plan.

a) Write an equation to represent this situation. **(1 mark)**

Equation: _____

b) Solve your equation to find how many months Zara had the plan. **(2 marks)**

Working: _____

_____

_____

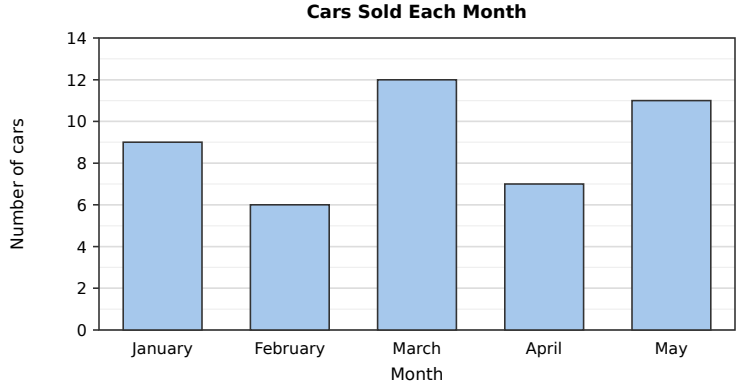
m = _____

Section B – Investigating Data (21 marks)

Suggested time: 20 minutes. Show all working.

Question 7 – Column graph **(2 marks)**

The column graph shows the number of cars sold by a car yard each month.



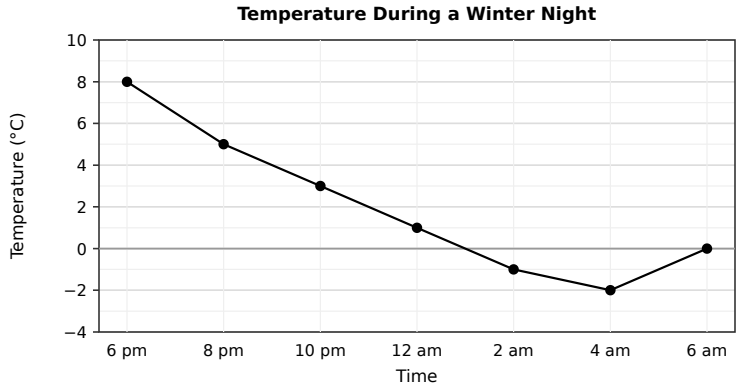
a) In which month were the most cars sold? _____ **(1 mark)**

b) How many cars were sold altogether in the five months? _____ **(1 mark)**

Working: _____

Question 8 – Line graph **(2 marks)**

The line graph shows the temperature in Canberra during a winter night.



a) At what time was the temperature the lowest? _____ **(1 mark)**

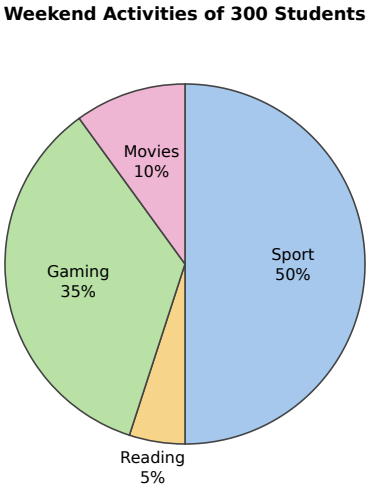
b) By how many degrees did the temperature fall between 6 pm and 4 am? **(1 mark)**

Working: _____

Answer: _____

Question 9 – Sector (pie) graph **(3 marks)**

300 students were asked how they like to spend their weekend. The results are shown in the sector graph.



a) How many students chose movies? **(1 mark)**

Working: _____

Answer: _____

b) How many more students chose gaming than reading? **(2 marks)**

Working: _____

_____

_____

Answer: _____

Question 10 – Stem-and-leaf plot **(2 marks)**

The test marks (out of 80) of 13 students are shown in the stem-and-leaf plot.

Stem	Leaf
4	5 8
5	0 2 2 7
6	1 3 4 6 9
7	0 4

Key: 5 | 2 means 52

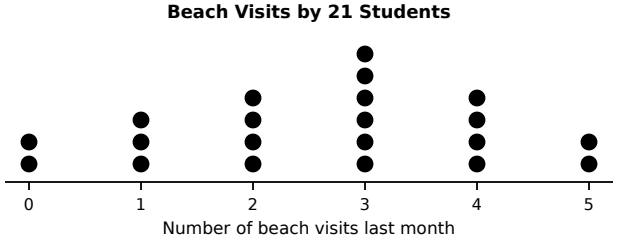
a) Find the median. _____ **(1 mark)**

Working: _____

b) How many students scored at least 64? _____ **(1 mark)**

Question 11 – Dot plot **(2 marks)**

Twenty-one students were asked how many times they went to the beach last month. The results are shown in the dot plot.



a) What is the mode? _____ **(1 mark)**

b) How many students went to the beach more than 3 times? _____ **(1 mark)**

Question 12 – Frequency table **(2 marks)**

A hockey team recorded the number of goals it scored in each game of the season. The results are shown in the frequency table.

Goals scored	0	1	2	3	4
Frequency	5	8	6	4	2

a) How many games did the team play altogether? _____ **(1 mark)**

Working: _____

b) What is the mode? _____ **(1 mark)**

Question 13 – Mean, median, mode and range **(5 marks)**

The ages of eight people in a choir are shown below.

23, 19, 27, 19, 31, 25, 19, 29

a) Calculate the mean. **(2 marks)**

Working: _____

_____

_____

Mean = _____

b) Find the median. _____ **(1 mark)**

Working: _____

c) Find the mode. _____ **(1 mark)**

d) Find the range. _____ **(1 mark)**

Working: _____

Question 14 – Outliers **(3 marks)**

The number of minutes that six students took to finish a puzzle is shown below.

42, 45, 40, 44, 8, 43

a) Identify the outlier in this data set. _____ **(1 mark)**

b) Calculate the mean of the data, including the outlier. **(1 mark)**

Working: _____

_____

Mean = _____

c) Does the outlier make the mean higher or lower than most of the data values? Explain your answer. **(1 mark)**

Working: _____

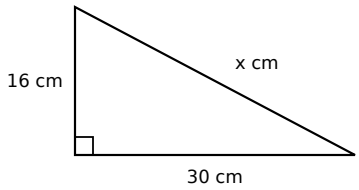
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Section C – Mixed Topics (21 marks)

Suggested time: 20 minutes. Show all working.

Question 15 – Pythagoras' Theorem **(4 marks)**

a) Find the length of the hypotenuse, *x*, in the triangle below. **(2 marks)**



Working: _____

_____

_____

x = _____

b) A triangle has side lengths 5 cm, 11 cm and 12 cm. Is it a right-angled triangle? Show working. **(2 marks)**

Working: _____

_____

_____

_____

Question 16 – Decimals **(2 marks)**

a) Calculate 0.7 × 0.08 _____ **(1 mark)**

Working: _____

_____

b) Calculate 8.1 ÷ 0.09 _____ **(1 mark)**

Working: _____

_____

Question 17 – Powers and roots **(2 marks)**

a) Evaluate 6^3 _____ **(1 mark)**

Working: _____

b) Evaluate √196 _____ **(1 mark)**

Working: _____

Question 18 – Surds **(3 marks)**

a) Circle the surds: √1 √45 √8 √121 **(1 mark)**

b) Simplify √80 **(2 marks)**

Working: _____

_____

_____

Answer: _____

Question 19 – Algebra **(3 marks)**

a) Simplify $3x^2 + 5x - x^2 + 2x$ _____ **(1 mark)**

Working: _____

b) Expand $-2(x - 6)$ _____ **(1 mark)**

Working: _____

c) Evaluate $4m^2 + 3n$ when $m = -2$ and $n = 5$. _____ **(1 mark)**

Working: _____

_____

Question 20 – Area and volume **(3 marks)**

a) Find the area of a trapezium with parallel sides 5.5 cm and 8.5 cm and height 6 cm. **(2 marks)**

Formula: Area = ½ × (a + b) × h, where a and b are the parallel sides and h is the perpendicular height.

Working: _____

_____

_____

Answer: _____

b) Find the volume of a rectangular prism that is 15 cm long, 4 cm wide and 2.5 cm high. **(1 mark)**

Working: _____

_____

Answer: _____

Question 21 – Linear relationships **(2 marks)**

Complete the table of values for the rule $y = -2x + 1$.

x	−1	0	1	2
y				

Question 22 – Rates and ratios **(2 marks)**

a) Write the ratio 45 : 60 in simplest form. _____ **(1 mark)**

Working: _____

_____

b) A bus travels 117 km in 1.5 hours. Find its average speed in km/h. **(1 mark)**

Working: _____

_____

Answer: _____

END OF EXAMINATION PAPER